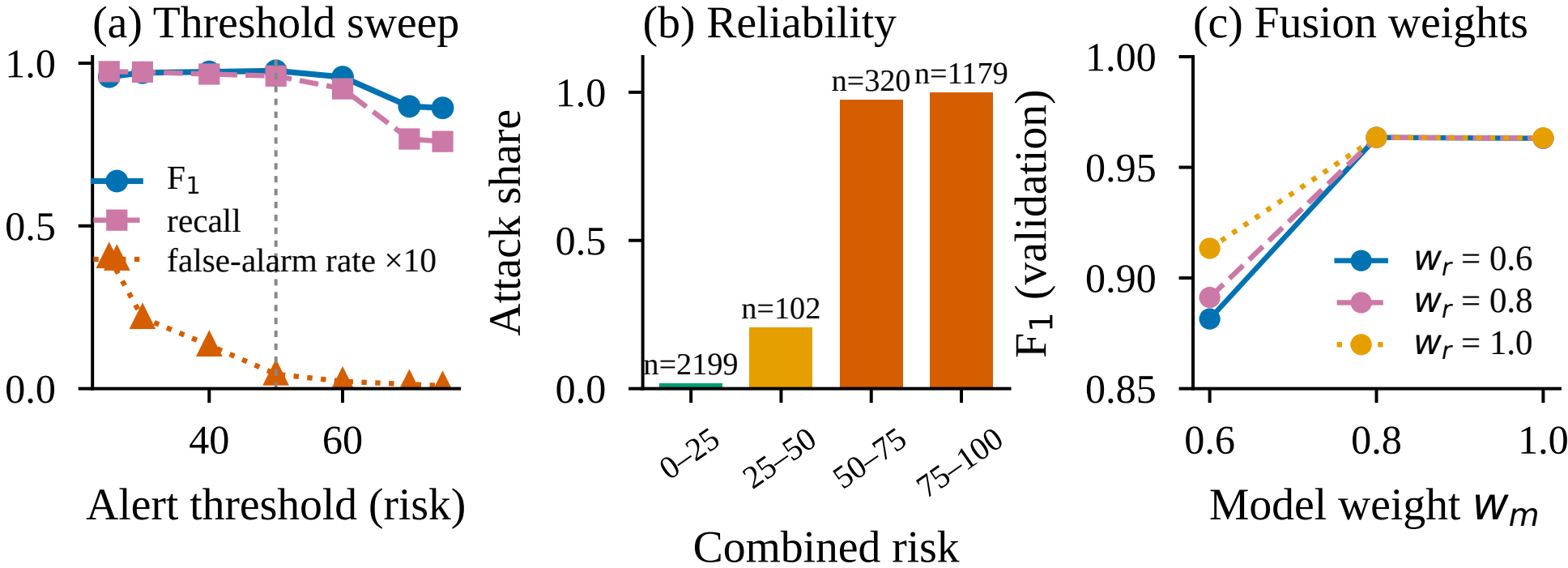
(a) Threshold sweep
1.0
0.5
0.0
F1
recall
false-alarm rate ×10
40
60
Alert threshold (risk)
(b) Reliability
Attack share
1.0
0.5
0.0
n=2199
n=102
n=320
n=1179
0–25
25–50
50–75
75–100
Combined risk
F1 (validation)
(c) Fusion weights
1.00
0.95
0.90
0.85
w_r = 0.6
w_r = 0.8
w_r = 1.0
0.6
0.8
1.0
Model weight w_m